

Morgan Baker

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Professional Experience

Computer Science TA

December 2021 - April 2025

Oregon State University, Remote

- Evaluated Python data structures and algorithms assignments by running and debugging student code to identify logic and syntax errors and improve code functionality.
 - Ran test cases and analyzed results to assess student code against given requirements, offering detailed, actionable feedback for improvement.
 - Guided students through debugging strategies during office hours, helping them trace errors, understand key concepts, and develop effective problem-solving skills.
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Project Experience

Capstone: Crowd-Sourced Travel Planner Web App

Oct 2023 - June 2024

- Implemented React.js frontend for intuitive and responsive user interface.
- Engineered the front end using React.js, resulting in an intuitive and responsive user interface.
- Developed backend services with Node.js, Express, and MongoDB to ensure seamless data storage and retrieval.
- Implemented authentication with Firebase, trip management, and itinerary planning features.
- Utilized Git for version control and conducted code reviews to maintain codebase quality.

Pixel Boost Web Application

April 2025 - August 2025

- Developed a full-stack application using FastAPI (Python) with MongoDB (beanie) and React (Typescript) to simulate real-life status tracking (hunger, thirst, etc.).
- Containerized application with Docker and deployed to AWS EC2 via GitHub Actions.
- Leveraged OAuth2 and JWT (cookies) for user authentication.
- Implemented a responsive UI based on Figma designs, for visual consistency and usability.

Resume Optimizer Web App

May 2025 - July 2025

- Built a full-stack AI application using the Gemini API, FastAPI (Python) and React to generate resumes tailored to a job description.
- Leveraged LLM prompting strategies (few-shot, text classification) to classify resume sections and generate content that highlights relevant skills.
- Integrated file parsing, PDF generation, and dynamic UI.

Supervised Project: Mobile Status Tracking App

Jan 2025 - April 2025

- Developed a React Native mobile application to simulate real-life status tracking (hunger, hydration, energy).
 - Integrated state management to improve performance and handle real-time status decay logic to mimic real-world needs.
 - Designed a pixel art style user interface using React Native components.
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Skills

- Languages: Python, Typescript, JavaScript, C/C++, HTML/CSS, Jinja2
 - Technologies: React, Git, Docker, AWS, Google Cloud Platform (GCP), FastAPI, Flask, Django, Node.js, Express, MongoDB, SQL, MySQL, OAuth2, Datastore, React Native, Kotlin
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Education

Bachelor of Science Computer Science

Aug 2021 - April 2025

Oregon State University. GPA: 3.6

I am excited to apply for the Applications Programmer Associate role at the University of Michigan. As a recent graduate in Computer Science from Oregon State University, I bring a strong foundation in software development and a passion for learning new technologies.

During my time at Oregon State University I was a TA for an Intro to Computer Science class based in Python which helped me strengthen my problem-solving, debugging, and communication skills. In my senior year I did a term long project where I developed a mobile application to learn React Native. My most recent project focused on learning the Python framework FastAPI and familiarizing myself with CI/CD pipelines through Docker, AWS, and Github Actions. These experiences have sharpened my ability to quickly adopt unfamiliar tools and frameworks.

I am confident that my technical foundation, eagerness to grow, and commitment to delivering quality software would allow me to excel in this role.

Thank you for your time and consideration. I look forward to the possibility of contributing to the University of Michigan's success.